

| DAY | FN/AN | PATTERN NUMBER | PATTERN              |
|-----|-------|----------------|----------------------|
| 1   | FN    | 1              | Prefix Sum           |
| 1   | FN    | 2              | Element Replace      |
| 1   | FN    | 3              | Reverse / Sort       |
| 1   | FN    | 4              | Custom Array         |
| 1   | FN    | 5              | Conditional Check    |
| 1   | FN    | 6              | String Manipulation  |
| 1   | AN    | 7              | Character Count      |
| 1   | AN    | 8              | String Equivalence   |
| 1   | AN    | 9              | Counting             |
| 1   | AN    | 10             | Matrix Traversal     |
| 2   | FN    | 1              | Frequency Count      |
| 2   | FN    | 2              | Array Filtering      |
| 2   | FN    | 3              | Even-Odd Logic       |
| 2   | FN    | 4              | Maximum Element      |
| 2   | FN    | 5              | String Join/Split    |
| 2   | FN    | 6              | 2D Grid Basics       |
| 2   | AN    | 7              | Comparison Counting  |
| 2   | AN    | 8              | String Checking      |
| 2   | AN    | 9              | Bit Logic            |
| 2   | AN    | 10             | Min/Max Position     |
| 3   | FN    | 1              | Digit Analysis       |
| 3   | FN    | 2              | Unique Values        |
| 3   | FN    | 3              | String Rebuild       |
| 3   | FN    | 4              | Set Usage            |
| 3   | FN    | 5              | Indexed Access       |
| 3   | FN    | 6              | Odd/Even Count       |
| 3   | AN    | 7              | Range Conditions     |
| 3   | AN    | 8              | Position Mapping     |
| 3   | AN    | 9              | Loop + Count         |
| 3   | AN    | 10             | Grid Logic           |
| 4   | FN    | 1              | Palindromes          |
| 4   | FN    | 2              | Array Rotation       |
| 4   | FN    | 3              | Vowel/Char Check     |
| 4   | FN    | 4              | Sorted Array Action  |
| 4   | FN    | 5              | Repeat Elements      |
| 4   | FN    | 6              | Frequency by Index   |
| 4   | AN    | 7              | Pair Comparison      |
| 4   | AN    | 8              | String Length/Bounds |
| 4   | AN    | 9              | String Composition   |
| 4   | AN    | 10             | Reshape/Transform    |
| 5   | FN    | 1              | Reverse              |
| 5   | FN    | 2              | Matrix Basics        |
| 5   | FN    | 3              | Vowel Logic          |
| 5   | FN    | 4              | Array Compare        |
| 5   | FN    | 5              | String Equivalence   |
| 5   | FN    | 6              | Subarray Basics      |
| 5   | AN    | 7              | Encoding Logic       |
| 5   | AN    | 8              | Sorting Logic        |
| 5   | AN    | 9              | Min/Max              |

|   |    |    |                  |
|---|----|----|------------------|
| 5 | AN | 10 | Count Conditions |
|---|----|----|------------------|

| Column    |
|-----------|
| PATTERN   |
| LEARN     |
| TRY       |
| SOLVE     |
| TAKE AWAY |

| LEARN                                                             | TRY                                                             |
|-------------------------------------------------------------------|-----------------------------------------------------------------|
| 1480. Running Sum of 1d Array                                     | 1672. Richest Customer Wealth                                   |
| 1299. Replace Elements with Greatest on Right Side                | 1389. Create Target Array in the Given Order                    |
| 344. Reverse String                                               | 977. Squares of a Sorted Array                                  |
| 1920. Build Array from Permutation                                | 1313. Decompress Run-Length Encoded List                        |
| 1342. Number of Steps to Reduce a Number to Zero                  | 1450. Number of Students Doing Homework at a Given Time         |
| 709. To Lower Case                                                | 1108. Defanging an IP Address                                   |
| 771. Jewels and Stones                                            | 389. Find the Difference                                        |
| 1662. Check If Two String Arrays are Equivalent                   | 1678. Goal Parser Interpretation                                |
| 1512. Number of Good Pairs                                        | 1365. How Many Numbers Are Smaller Than the Current Number      |
| 1572. Matrix Diagonal Sum                                         | 566. Reshape the Matrix                                         |
| 242. Valid Anagram                                                | <b>387. First Unique Character in a String</b>                  |
| 804. Unique Morse Code Words                                      | 929. Unique Email Addresses                                     |
| 922. Sort Array By Parity II                                      | 905. Sort Array by Parity                                       |
| 414. Third Maximum Number                                         | 1984. Minimum Difference Between Highest and Lowest of K Scores |
| 557. Reverse Words in a String III                                | 345. Reverse Vowels of a String                                 |
| <b>1572. Matrix Diagonal Sum</b>                                  | 1588. Sum of All Odd Length Subarrays                           |
| 1351. Count Negative Numbers in a Sorted Matrix                   | 1431. Kids With the Greatest Number of Candies                  |
| 1844. Replace All Digits with Characters                          | 2108. Find First Palindromic String in the Array                |
| 191. Number of 1 Bits                                             | 231. Power of Two                                               |
| 1295. Find Numbers with Even Number of Digits                     | 1523. Count Odd Numbers in an Interval Range                    |
| 7. Reverse Integer                                                | 9. Palindrome Number                                            |
| 217. Contains Duplicate                                           | 268. Missing Number                                             |
| <b>557. Reverse Words in a String III</b>                         | <b>344. Reverse String</b>                                      |
| <b>771. Jewels and Stones</b>                                     | <b>804. Unique Morse Code Words</b>                             |
| <b>1380. Lucky Numbers in a Matrix</b>                            | <b>1572. Matrix Diagonal Sum</b>                                |
| <b>1207. Unique Number of Occurrences</b>                         | 1913. Maximum Product Difference Between Two Pairs              |
| 1108. Defanging an IP Address                                     | <b>709. To Lower Case</b>                                       |
| 1470. Shuffle the Array                                           | <b>1389. Create Target Array in the Given Order</b>             |
| <b>1342. Number of Steps to Reduce a Number to Zero</b>           | 876. Middle of the Linked List                                  |
| <b>566. Reshape the Matrix</b>                                    | 485. Max Consecutive Ones                                       |
| 125. Valid Palindrome                                             | 680. Valid Palindrome II                                        |
| 189. Rotate Array                                                 | 283. Move Zeroes                                                |
| <b>345. Reverse Vowels of a String</b>                            | <b>1108. Defanging an IP Address</b>                            |
| 977. Squares of a Sorted Array                                    | 1002. Find Common Characters                                    |
| <b>1207. Unique Number of Occurrences</b>                         | 961. N-Repeated Element in Size 2N Array                        |
| 1380. Lucky Numbers in a Matrix                                   | <b>1450. Number of Students Doing Homework at a Given Time</b>  |
| 1572. Matrix Diagonal Sum                                         | 1346. Check If N and Its Double Exist                           |
| <b>2114. Max Number of Words Found</b>                            | <b>1832. Check if the Sentence Is Pangram</b>                   |
| <b>1678. Goal Parser Interpretation</b>                           | 1662. Check If Two String Arrays are Equivalent                 |
| <b>1260. Shift 2D Grid</b>                                        | <b>118. Pascal's Triangle</b>                                   |
| 344. Reverse String                                               | 557. Reverse Words in a String III                              |
| <b>1572. Matrix Diagonal Sum</b>                                  | <b>566. Reshape the Matrix</b>                                  |
| <b>345. Reverse Vowels of a String</b>                            | 1119. Remove Vowels from a String                               |
| <b>1365. How Many Numbers Are Smaller Than the Current Number</b> | 1385. Find the Distance Value                                   |
| <b>1662. Check If Two String Arrays are Equivalent</b>            | <b>929. Unique Email Addresses</b>                              |
| <b>53. Maximum Subarray</b>                                       | 121. Best Time to Buy and Sell Stock                            |
| <b>1313. Decompress Run-Length Encoded List</b>                   | 1920. Build Array from Permutation                              |
| <b>905. Sort Array by Parity</b>                                  | <b>922. Sort Array by Parity II</b>                             |
| <b>414. Third Maximum Number</b>                                  | <b>1984. Minimum Difference Between Scores</b>                  |

**1281. Subtract Product and Sum****1295. Find Numbers with Even Digits****Meaning**

Core concept behind the problem

Grasp the pattern through guided learning

Apply the concept with focused practice

Crack problems using both pattern and practice

Key insight or template gained

| SOLVE                                                           | TAKE AWAY                                          |
|-----------------------------------------------------------------|----------------------------------------------------|
| 1732. Find the Highest Altitude                                 | All involve accumulating values with simple loops. |
| 1380. Lucky Numbers in a Matrix                                 | Focus on updating elements using traversal.        |
| 905. Sort Array by Parity                                       | All use two-pointer tricks or array modification.  |
| 1720. Decode XORed Array                                        | All need building new array from rules.            |
| 1207. Unique Number of Occurrences                              | Condition-based counting with simple logic.        |
| 1832. Check if the Sentence Is Pangram                          | All manipulate characters or strings.              |
| 387. First Unique Character in a String                         | Counting frequency of characters.                  |
| 1528. Shuffle String                                            | Rearranging or interpreting strings.               |
| 1748. Sum of Unique Elements                                    | Counting, comparing or checking occurrences.       |
| 1266. Minimum Time Visiting All Points                          | Simple 2D traversal and reshaping.                 |
| 383. Ransom Note                                                | All use character frequency/hash map.              |
| 2215. Find the Difference of Two Arrays                         | Filter + transform + store in set.                 |
| 2154. Keep Multiplying Found Values by Two                      | All based on even/odd or divisibility.             |
| 2194. Cells in a Range on an Excel Sheet                        | All require finding max/min or range.              |
| <b>709. To Lower Case</b>                                       | Manipulate or split/join string parts.             |
| 1260. Shift 2D Grid                                             | Looping through 2D grids consistently.             |
| 2114. Maximum Number of Words Found in Sentences                | Compare and count values per condition.            |
| 2124. Check if All A's Appears Before All B's                   | Check and modify characters.                       |
| 1281. Subtract the Product and Sum of Digits of an Integer      | Basic number and bit operations.                   |
| 2235. Add Two Integers                                          | Count & locate simple numeric property.            |
| 66. Plus One                                                    | Work with digits using mod/div.                    |
| 136. Single Number                                              | Set/map usage for uniqueness.                      |
| 451. Sort Characters by Frequency                               | Basic reversal and sorting of characters.          |
| 202. Happy Number                                               | All use set to track unique data.                  |
| 118. Pascal's Triangle                                          | Use index rules to access elements.                |
| <b>1365. How Many Numbers Are Smaller Than the Current N</b>    | All use counting and comparison logic.             |
| 1752. Check if Array Is Sorted and Rotated                      | Range-based checks on values or chars.             |
| 1576. Replace All ?'s to Avoid Consecutive Repeating Characters | Map values based on given index rules.             |
| <b>1450. Number of Students Doing Homework at a Given Time</b>  | Use while/for with counters.                       |
| 463. Island Perimeter                                           | Basic grid traversal.                              |
| 409. Longest Palindrome                                         | Understand reverse and character frequency.        |
| 1051. Height Checker                                            | Shift elements while preserving order.             |
| <b>1528. Shuffle String</b>                                     | Operate conditionally on characters.               |
| 1089. Duplicate Zeros                                           | Operate based on sorted structure.                 |
| 136. Single Number                                              | Analyze repeated values.                           |
| 1748. Sum of Unique Elements                                    | Index-based frequency logic.                       |
| <b>1512. Number of Good Pairs</b>                               | Check and count value relationships.               |
| <b>709. To Lower Case</b>                                       | Work with string size and checks.                  |
| 2119. A Number After a Double Reversal                          | Construct/validate new strings.                    |
| <b>832. Flipping an Image</b>                                   | Create new 2D shape or logic.                      |
| 917. Reverse Only Letters                                       | Same reverse logic in new forms.                   |
| 1260. Shift 2D Grid                                             | Matrix transformation and loop.                    |
| <b>1832. Check if the Sentence Is Pangram</b>                   | Vowel-based filtering/logic.                       |
| 349. Intersection of Two Arrays                                 | Compare elements across arrays.                    |
| <b>709. To Lower Case</b>                                       | String normalization and equality check.           |
| 724. Find Pivot Index                                           | Use prefix, suffix, or current sum.                |
| <b>1720. Decode XORed Array</b>                                 | Encode/decode based on rules.                      |
| 1122. Relative Sort Array                                       | Sort with conditions.                              |
| 628. Max Product of 3 Numbers                                   | Finding top/max/min with sort/loop.                |

**191. Number of 1 Bits**

Count digits, bits, or length.



GRAPH



## STRUCTURE AND ALGORITHM

### PROBLEM

LC 19

LC 203

LC 876

LC 237

LC 2095

Delete last occurrence of an item from linked list-GFG

LC 83

LC 82

LC 206

LC 2074

LC 234

LC 24

LC 61

LC 92

LC 25

LC 2

LC 445

LC 369

LC 21

LC 23

LC 160

LC 143

LC 138

LC 328

LC 86

LC 141

LC 142

LC 94

LC 144

LC 145

GFG Sum of Binary Tree

GFG Hight of Binary Tree

LC 104

LC 543

LC 100

LC 112

LC 101

LC 102

LC 103

LC 107

LC 199

LC 105

GFG Print adjacency list

GFG DFS of Graph

GFG BFS of graph

LC 200

LC 695

LC 419

LC 2658

LC 3619

LC 1905

LC 1992

LC 130

LC 733

LC 1905

LC 1922

LC 417

LC 547

LC 1020

LC 1791

LC 1971

LC 133

LC 127

# COMPETITIVE PROGRAMS PROBLEMS

| CODE NO      | CODE NAME                      |
|--------------|--------------------------------|
| 1899A (CF)   | A. Game with Integers          |
| 1878A (CF)   | A. How Much Does Daytona Cost? |
| 1901A (CF)   | A. Line Trip                   |
| 1900A (CF)   | A. Cover in Water              |
| 1903A (CF)   | A. Halloumi Boxes              |
| 1890A (CF)   | A. Doremy's Paint 3            |
| 1877A (CF)   | A. Goals of Victory            |
| 1896A (CF)   | A. Jagged Swaps                |
| 1859A (CF)   | A. United We Stand             |
| 1866A (CF)   | A. Ambitious Kid               |
| 1862B (CF)   | B. Sequence Game               |
| HACKER RANK  | Array Manipulation             |
| HACKER EARTH | Bob and string                 |
| HACKER EARTH | Xsquare and Double String      |
| 1913B (CF)   | B. Swap and Delete             |
| 1883C (CF)   | C. Raspberries                 |
| 1791D (CF)   | D. Distinct Split              |
| 1765M (CF)   | M. Minimum LCM                 |
| 1715B (CF)   | B. Beautiful Array             |
| 1725B (CF)   | B. Basketball Together         |
| 1859B (CF)   | B. Olya and Game with Arrays   |
| 1849B (CF)   | B. Monsters                    |
| 1840C (CF)   | C. Ski Resort                  |
| 1831B (CF)   | B. Array Merging               |
| 1632B (CF)   | B. Roof Construction           |